

Worksheet W9: e^x — The Super-Function

"The function that equals its own derivative."

Chapter 9 · 10 Problems · Name: _____ Date: _____

e^x is its own derivative: $\frac{d}{dx}(e^x) = e^x$ | $\int e^x dx = e^x + C$

Problem 1

Question: What is the number e approximately equal to?

💡 Hint: $e \approx 2.718...$ It's an irrational number.

Problem 2

Question: What is the **derivative** of e^x ?

💡 Hint: e^x is its own derivative.

Problem 3

Question: What is the **integral** of e^x ?

💡 Hint: e^x is its own integral too. Don't forget $+C$.

Problem 4

Find the derivative of $f(x) = e^{2x}$.

💡 Hint: The derivative of e^{kx} is $k \times e^{kx}$. Here $k = 2$.

Problem 5

Find the derivative of $f(x) = 3e^x$.

💡 Hint: The coefficient stays in front.

Problem 6

Find the derivative of $f(x) = 4e^{3x}$.

💡 Hint: Multiply the coefficient by k . $4 \times 3 = 12$.

Problem 7

Find the integral of $2e^{2x}$.

💡 Hint: $\int e^{kx} dx = \frac{1}{k} e^{kx} + C$.

Problem 8

Question: What does the "Magic Bank Account" illustrate about e^x ?

💡 Hint: Continuous growth at a rate equal to the current amount.

Problem 9

Question: What is the derivative cycle for e^x ?

💡 Hint: The derivative of e^x is e^x . The derivative of that is also e^x .

Problem 10

Question: Why is e^x called the "**super-function**"?

💡 *Hint: It is its own derivative AND its own integral.*